

TRONDHEIM VOLL, Norway

WMO: 012570

Lat: 63.4106N

Lon: 10.4539E

Elev: 132

StdP: 99.76

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.1	-11.8	-18.0	0.8	-12.4	-16.2	0.9	-10.2	10.1	0.8	9.2	1.7	2.2	210	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	24.8	16.4	22.9	15.6	21.0	14.9	17.6	23.0	16.5	21.4	15.5	19.9	2.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	11.0	19.9	14.2	10.2	18.6	13.3	9.7	17.5	49.8	23.0	46.6	21.5	43.7	20.0	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	6.9	5.9	-16.2	27.5	3.1	2.0	-18.4	29.0	-20.2	30.1	-21.9	31.2	-24.1	32.7		
(5)			-16.6	19.4	2.9	1.4	-18.7	20.4	-20.4	21.2	-22.0	22.0	-24.1	23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-1.6	-1.3	0.7	5.0	9.0	12.1	15.2	14.4	10.9	6.1	1.9	-0.3
	DBStd	7.06	4.73	4.73	4.01	3.29	4.00	3.31	3.46	2.96	2.94	3.40	4.19	4.70
	HDD10.0	1919	359	315	288	156	69	14	1	2	24	129	244	318
	HDD18.3	4508	617	548	547	402	291	189	111	127	224	381	494	576
	CDD10.0	474	0	0	0	4	37	76	162	138	49	7	0	0
	CDD18.3	21	0	0	0	0	1	2	13	6	0	0	0	0
	CDH23.3	138	0	0	0	0	4	18	94	21	0	0	0	0
	CDH26.7	19	0	0	0	0	0	1	18	1	0	0	0	0
Wind	WSAvg	2.7	3.1	2.9	3.1	2.7	2.6	2.5	2.2	2.2	2.5	2.6	2.7	3.0
Precipitation	PrecAvg	835	67	59	59	48	42	67	66	61	107	102	70	87
	PrecMax	1235	138	116	200	90	92	136	113	116	197	174	163	328
	PrecMin	542	1	2	3	14	4	12	12	9	18	11	16	20
	PrecStd	204	36	31	45	21	24	32	30	26	43	50	42	71
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	8.8	10.6	18.2	23.6	25.8	28.9	25.5	21.0	16.6	11.8	9.1
		MCWB	4.2	4.9	5.4	10.1	14.5	15.7	19.5	16.4	13.7	11.4	8.1	5.9
	2%	DB	6.4	6.9	8.3	14.8	20.9	22.8	26.0	23.6	18.7	14.5	10.3	7.7
		MCWB	3.7	3.6	4.7	8.5	13.3	14.5	17.6	16.5	13.4	9.9	7.0	4.6
	5%	DB	5.5	5.6	7.2	12.5	18.5	20.1	23.5	21.6	16.9	12.7	8.6	6.6
		MCWB	3.1	2.8	4.1	7.2	12.3	13.8	16.3	15.6	12.3	9.0	5.7	3.9
	10%	DB	4.3	4.6	6.0	10.1	15.9	18.0	21.5	19.7	15.4	10.9	7.3	5.5
		MCWB	2.4	2.3	3.1	6.0	10.8	12.9	15.5	14.8	11.6	7.9	4.8	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	5.4	6.9	10.9	15.7	16.9	20.3	18.3	15.2	12.1	9.0	6.9
		MCDB	6.4	7.6	8.3	16.9	22.4	22.8	27.0	23.6	19.3	16.0	11.3	7.8
	2%	WB	4.5	4.3	5.6	9.1	13.9	15.4	18.4	17.1	13.9	10.3	7.5	5.5
		MCDB	5.8	6.0	7.6	13.7	19.5	21.2	24.8	21.9	17.2	13.6	9.7	7.0
	5%	WB	3.4	3.4	4.4	7.7	12.6	14.3	17.0	16.2	13.0	9.4	6.2	4.2
		MCDB	4.9	5.2	6.7	11.5	17.8	19.0	22.3	20.7	16.0	12.2	8.1	6.0
	10%	WB	2.5	2.4	3.4	6.5	11.0	13.2	15.9	15.1	12.1	8.3	5.0	3.3
		MCDB	4.1	4.3	5.6	9.7	16.0	17.4	20.5	18.8	14.8	10.7	7.0	5.2
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.8	5.9	6.8	7.2	7.0	7.3	7.0	6.1	5.3	4.5	4.7
		MCDBR	4.6	4.9	6.7	10.2	11.5	11.0	11.2	10.1	8.3	6.9	6.0	5.5
		MCWBR	3.3	3.6	4.4	5.7	5.9	5.4	5.1	4.9	4.6	4.3	4.1	4.2
	5% WB	MCDBR	4.3	4.6	6.1	9.0	10.9	9.8	10.6	9.2	7.9	6.1	5.8	5.3
		MCWBR	3.4	3.7	4.5	5.4	5.9	5.2	5.2	4.8	4.7	4.1	4.4	4.3
Clear-Sky Solar Irradiance	taub	0.223	0.285	0.314	0.341	0.344	0.345	0.354	0.342	0.322	0.300	0.365	0.355	
	taud	2.043	2.237	2.290	2.372	2.447	2.478	2.474	2.503	2.545	2.457	2.732	2.655	
	Ebn at Noon	393	650	780	841	871	876	856	836	785	645	375	218	
	Edh at Noon	29	59	83	95	97	96	94	84	66	48	26	17	
All-Sky Solar Radiation	RadAvg	0.16	0.69	1.95	3.74	4.58	4.77	4.62	3.55	2.13	0.94	0.25	0.08	
	RadStd	0.02	0.10	0.28	0.30	0.30	0.48	0.47	0.41	0.22	0.13	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-143	-189	+67	N/A

Nomenclature: See separate page